

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
import kagglehub
```

```
path3 = kagglehub.dataset_download("adelanseau/taxi-trips-chicago-2024")
print("Path to dataset files:", path3)
```

Downloading to /root/.cache/kagglehub/datasets/adelanseau/taxi-trips-chicago-2024/1.archive...
100%|██████████| 111M/111M [00:03<00:00, 33.5MB/s]Extracting files...

Path to dataset files: /root/.cache/kagglehub/datasets/adelanseau/taxi-trips-chicago-2024/versions/1

```
import os
print(os.listdir(path3))
```

```
['Taxi_Trips_-_2024_20240408.csv']
```

```
data3 = pd.read_csv("/root/.cache/kagglehub/datasets/adelanseau/taxi-trips-chicago-2024/versions/1/Taxi_Trips_-_2024_20240408.csv")
```

```
data3 = data3[['Trip Miles', 'Trip Total']]
```

```
data3.shape
```

```
(865247, 2)
```

```
data3.dropna(inplace=True)
data3.isnull().sum()
```

```

0
Trip Miles    0
Trip Total    0
dtype: int64
```

```
data3.head()
```

	Trip Miles	Trip Total
0	0.09	38.75
1	3.00	12.00
2	5.84	26.70
3	13.36	47.10
4	6.13	22.82

```
x3 = data3.iloc[:, :1].values
y3 = data3.iloc[:, 1:2].values
print(x3)
print(y3)
```

```

[[ 0.09]
 [ 3.   ]
 [ 5.84]
 ...
 [17.44]
 [ 3.07]
 [ 1.49]]
[[38.75]
 [12.   ]
 [26.7 ]
 ...
 [66.3 ]
 [26.   ]
 [12.25]]
```

```
x3_train,x3_test,y3_train,y3_test = train_test_split(x3,y3,test_size = 0.2,random_state = 42)
```

```
model3 = LinearRegression()
model3.fit(x3_train, y3_train)
```

```
y3_pred = model3.predict(x3_test)
```

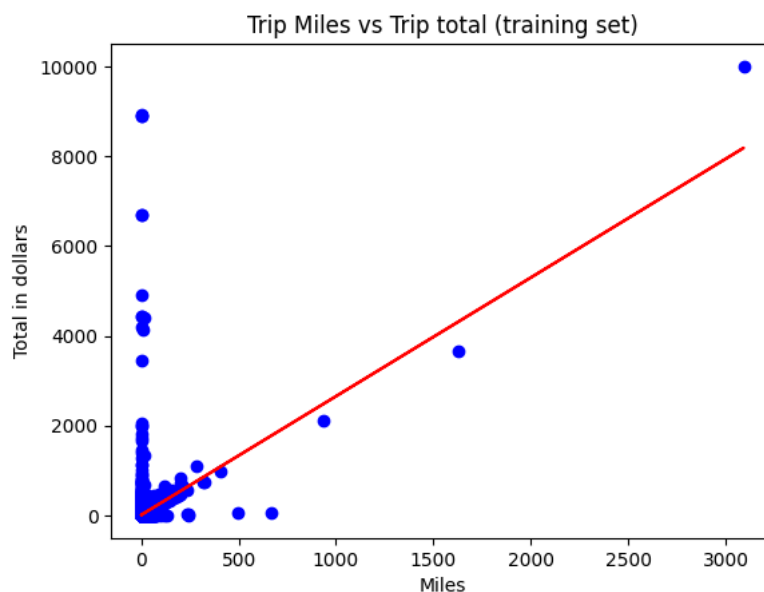
```
model3.score(x3_test,y3_test)
```

```
0.1719224457128684
```

```
print(y3_test)
print(y3_pred)
```

```
[[31.25]
 [24.98]
 [ 8.75]
 ...
 [45.25]
 [43.5 ]
 [50.4 ]]
[[42.1919655 ]
 [28.15969042]
 [12.27758623]
 ...
 [ 9.81995614]
 [48.63994124]
 [48.11141863]]
```

```
plt.scatter(x3_train, y3_train, color = 'blue')
plt.plot(x3_train, model3.predict(x3_train), color = 'red')
plt.title('Trip Miles vs Trip total (training set)')
plt.xlabel('Miles')
plt.ylabel('Total in dollars')
plt.show()
```



```
print(x3_train.shape)
print(x3_test.shape)
print(model3.coef_)
print(model3.intercept_)
```

```
(690556, 1)
(172640, 1)
[[2.64261301]]
[9.81995614]
```

```
plt.scatter(x3_test, y3_test, color = 'blue')
plt.plot(x3_train, model3.predict(x3_train), color = 'red')
plt.title('Trip Miles vs Total (training set)')
plt.xlabel('Miles')
plt.ylabel('Total in dollars')
plt.show()
```

